

Switch statement structure

switch statement: another (easier) way express nested if-else statement

```
1  switch (integral-expression)
2  {
3      case integral-value 1: corresponding statements;
4                          break;
5      case integral-value 2: corresponding statements;
6                          break;
7      ...
8      default: corresponding statements;
9      //the default part is optional.
10 }
```

Switch statement: more details

- Only integer expression[★] can be used after keyword **switch**.
- Only integer value[★] can be used after keyword **case**.
- Cannot use string in switch statement in C++.

Switch Statement: more details II

- Statement **break**; means to jump out the switch-block (enclosed by { and }).
- In each case, there is a colon **:** after each value, enter **break**;
 - Statement **break**; **right before the closing }** can be omitted.
- **default**: means all the rest cases, it is optional.
- However, if **default**: is present, it must be put in the end of all other cases.

Warning: missing `break;` in not-the-last case

- What is the output when input is 6?

```
cout << "Enter an int: ";
int num;
cin >> num;

switch (num % 2)
{
    case 0: cout << num << " is even." << endl;

    case 1: //when num is positive odd int
    case -1: //when num is negative odd int
        cout << num << " is odd." << endl;
}
```

Fix of the above code

- What is output when input 6 in the following code?

```
cout << "Enter an int: ";
int num;
cin >> num;

switch (num % 2)
{
    case 0: cout << num << " is even." << endl;
            break;
    case 1: //when num is positive odd int
    case -1: //when num is negative odd int
            cout << num << " is odd." << endl;
}
```

Another fix

What is the output when input is 6?

```
cout << "Enter an int: ";
int num;
cin >> num;

switch (num % 2)
{
    case 0: cout << num << " is even." << endl;
            break;
    default: //num % 2 is not 0, it can be 1 or -1
            cout << num << " is odd." << endl;
}
```

Even or Odd: use if-else statement

- In this example, since there are only one if-else statement, we do not need to use switch statement.

```
cout << "Enter an int: ";  
int num;  
cin >> num;  
  
if (num % 2 == 0)  
    cout << num << " is even." << endl;  
else cout << num << " is odd." << endl;
```

Exercise

- Define a method, given numerical number (can be double type), return the corresponding letter grade.
- Use switch statement.